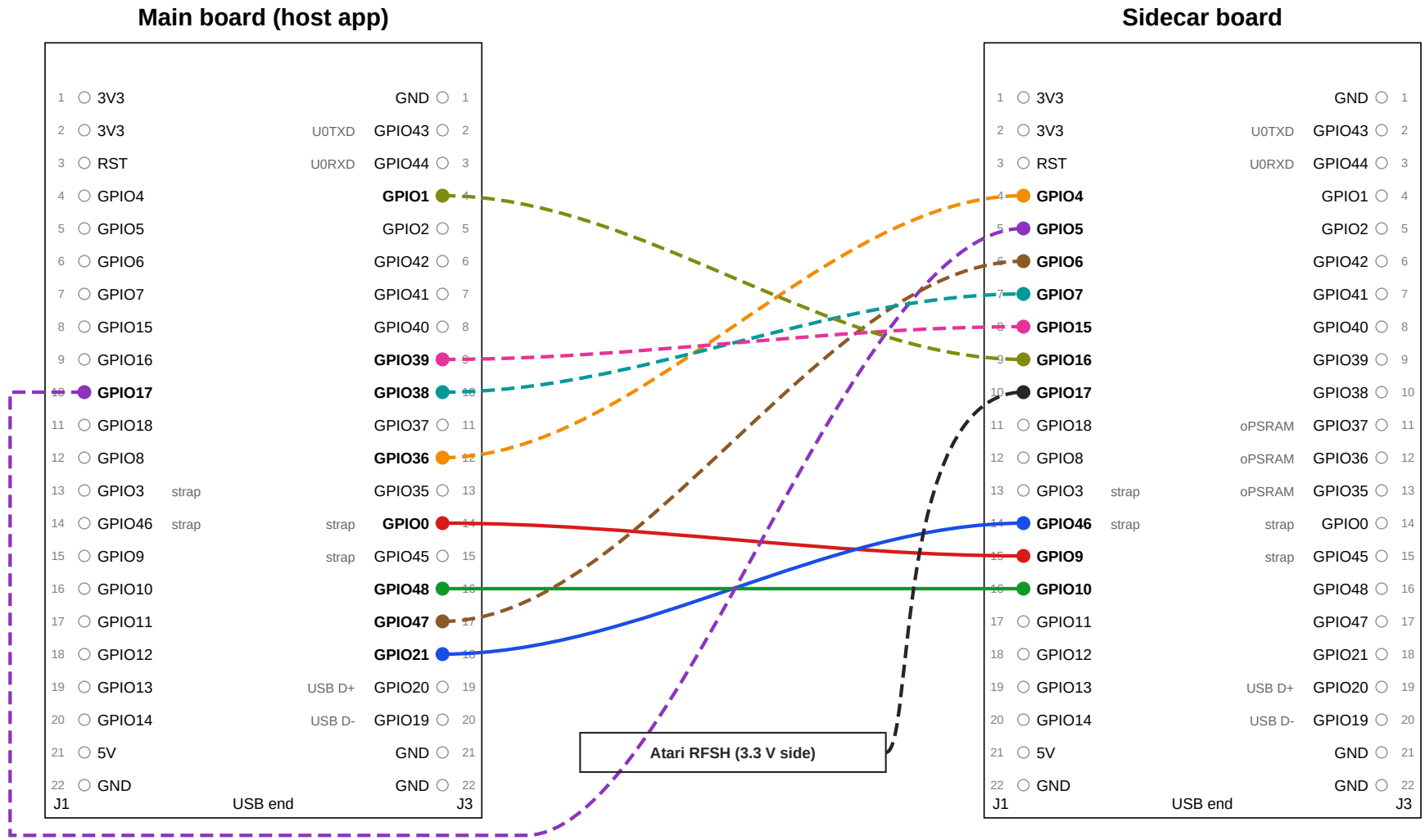
Main board <-> Sidecar wiring (ESP32-S3-DevKitC-1 pair)

Generated 2026-11-11 by docs/sidecar_wiring.py from its WIRES table. Ground and power come through the carrier board (not drawn). Header pin numbers: DevKitC-1, pin 1 away from the USB end.



Signal	From	To	Header (from -> to)	Status	Notes
Link TX	main GPIO0	Sidecar GPIO9	J3-14 -> J1-15	wired	host TX -> Sidecar RX; host GPIO0 also reaches PBI audio
Link RX	Sidecar GPIO46	main GPIO21	J1-14 -> J3-18	wired	Sidecar TX -> host RX; host GPIO21 is refresh_ in non-Sidecar builds
BCLK	main GPIO48	Sidecar GPIO10	J3-16 -> J1-16	wired	host drives the link clock
phi2	main GPIO36	Sidecar GPIO4	J3-12 -> J1-4	planned	Sidecar bus profiler: clock
R/W	main GPIO17	Sidecar GPIO5	J1-10 -> J1-5	planned	Sidecar bus profiler
extSel_	main GPIO47	Sidecar GPIO6	J3-17 -> J1-6	planned	Sidecar bus profiler; open drain on the host, keep it short
D0	main GPIO38	Sidecar GPIO7	J3-10 -> J1-7	planned	Sidecar bus profiler
D1	main GPIO39	Sidecar GPIO15	J3-9 -> J1-8	planned	Sidecar bus profiler
A0	main GPIO1	Sidecar GPIO16	J3-4 -> J1-9	planned	Sidecar bus profiler
RFSH	Atari RFSH	Sidecar GPIO17	- -> J1-10	planned	Sidecar bus profiler refresh_; 3.3 V side, never the 5 V bus line

Solid: wired now. Dashed: planned. Filled pin: has a wire. oPSRAM: octal-PSRAM pin on N8R8 modules.